

ASSIGNMENT # 01

SOFTWARE ENGINEERING

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SECTION:

Software Engineering

AFTERNOON (A)

UNIVERSITY LMS

A university Learning Management System (LMS) is a software platform used by universities to manage courses, lectures, assignments, quizzes, grades, announcements, and communication between students and teachers.

A. Purpose:

The main purpose of a University LMS is to provide online platform.

It allows students to:

- Register courses
- View their information
- View grades and feedback
- Receive announcements.

It allows teachers to:

- Take attendance
- Give grades and feedback
- Communicate with students.
- Create assignments and quizzes.

B. Users:

1. **Students** - access courses, submit assignments, take quizzes, and view grades.
2. **Teachers/Instructors** - manage courses, assignment quizzes, and grades.

3. **Administrators** — manage users, courses, permissions, and the overall system.

4. **Department staff** — manage academic information and course enrollment.

c. Stakeholders :

The stakeholders include :

1. Students
2. Teachers / Instructors
3. University administration
4. IT / System administrators
5. Departments and academic staff.
6. University management.

Stakeholders are people or organizations that are affected by or have an interest in LMS.

d. Functional Requirements:

Functional requirements describe what the system must do.

1. **User Authentication** — The LMS shall allow students, teachers, and administrators to log in using their authorized credentials.

2. **Course Management** — The system shall allow teachers and administrators to create, update, and manage courses and course materials.

3. **Assignment Submission** — The system shall allow students to upload and submit assignments before the deadline.

4. **Grade Management** — The system shall allow teachers to enter grades and students to view their grades.

E. Non-Functional Requirements:

Non-Functional requirements describe how well the system should perform.

1. **Performance:** The LMS should respond to normal user requests within a reasonable amount of time, even when many students are using it simultaneously.
2. **Security:** The system should protect student and teacher information using authentication, authorization, and secure data storage.
3. **Availability:** The LMS should be available whenever students and teachers need to access academic resources, with minimal downtime.
4. **Usability:** The interface should be simple easy to understand so that students and teachers can perform common tasks without difficulty.
5. **Scalability:** The system should be capable of supporting an increasing number of students, teachers, courses, and stored resources.

F. Risks:

1. **Security / Data Breach** - Unauthorized users could gain access to student records, grades, assignments, or personal information.

Impact - Loss of privacy and trust.

2. **System Downtime** - The LMS could become unavailable because of server problems, network

failures, or software errors.

Impact - Students may be unable to submit assignments or take online quizzes.

3. Data Loss - Important information such as grades, assignments or course materials could be lost because of hardware failures, software errors, or inadequate backups.

Impact - Academic records could be affected.

9. Quality Attributes:

- 1. Reliability** - The system should work correctly and consistently without frequent crashes or errors.
- 2. Security** - The system should ensure that only authorized users can access protected academic and personal information.
- 3. Usability** - The LMS should have a clear and user-friendly interface that makes it easy for students and teachers to use.

H. Where AI could be used

- 1. AI Chatbot** - answer common student questions about courses, deadlines, and university procedures.
- 2. Personalized Learning Recommendations** - recommend learning materials or topics to students based on their learning activity.
- 3. Automated Feedback** - AI could provide preliminary feedback on certain assignments and quizzes.

4. **Student Performance Analysis** - AI could identify students who may be struggling based on their academic activity and performance.

5. **Content Generation Assistance** - AI could help teachers create quiz questions, summaries or study materials.

I. What could go wrong if AI is used incorrectly?

1. **Incorrect information** - AI may provide wrong answers or misleading academic information.

2. **Unfair Grading** - If AI is incorrectly used for grading, it could give unfair or inaccurate grades.

3. **Privacy Problems** - Sensitive student information could be exposed or improperly processed.

4. **Bias** - An AI system may produce biased recommendations or evaluations if its data or design contain bias.

5. **Overdependence on AI** - Students may rely too heavily on AI instead of developing their own understanding and problem-solving skills.